Document Control

Title

Lidocaine Infusion Therapy for Chronic Neuropathic Pain Standard Operating Procedure

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Consulted with the following stakeholders	**Contact responsible for implementation and monitoring compliance:**
• Pain Consultants • Pain Nurses • Ward Manager Seamoor Unit	Chronic Pain Lead
	Education/ training will be provided by: Chronic Pain Team

Approval and Review Process
- Anaesthetic Lead Clinician

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Chronic Pain	Chronic, Pain, Persistent Pain, Lidocaine, Infusion, Therapy, Neuropathic

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1. Background

1.1. Neuropathic pain is a disorder that can be accompanied by severe pain that is both chronic and resistant to conventional therapies. Lidocaine is a local anaesthetic which blocks voltage gated sodium channels within neurons to block transmission of pain. It is commonly used as a local anaesthetic injection in the skin or associated tissues for surgery as well as in nerve blocks for anaesthesia.

1.2. However, thereThere is a substantial body of research which demonstrates the it also pain reducing value of Lidocaine administered as an intra-venous infusion for both acute and chronic, neuropathic pain. Patients in whom neuropathic pain has been resistant to other forms of treatment may gain benefit from lidocaine infusions.

1.3. The infusion regimens differ between contexts and also between institutions. This SOP specifically applies to infusions performed as a day case procedure on the authority of a Consultant Anaesthetist with an Interest in Pain Medicine.

2. Purpose

2.1. The Standard Operating Procedure (SOP) has been written to:

- Ensure that standard infusion doses are prescribed and administered in all patients with neuropathic pain requiring lidocaine intravenous infusion.
- Ensure that intravenous lidocaine infusions are administered in a safe manner, correctly prescribed and administered, using the correct equipment.
- Ensure that appropriate monitoring of patients using intravenous lidocaine infusions occurs and that actions in event of adverse reactions are described and equipment available.

3. Scope

3.1. This Standard Operating Procedure (SOP) relates to the following staff groups who may be involved in the assessment and delivery of pain relief:

- Registered nurses
- Operating Department Practitioners
- Medical staff – Anaesthetists with an interest in chronic pain

3.2. A Chronic Pain consultant should assess the patient in the outpatient clinic to assess and consent for treatment with an intravenous lidociane infusion as a day case procedure to be booked at a separate appointment.

4. Location

4.1. This Standard Operating Procedure regarding intravenous lidocaine infusions for chronic neuropathic pain can be implemented in the Seamoor unit where competent members of staff are available to undertake this role with the required levels of patient monitoring and standard resuscitation facilities are available.

4.2. Staff undertaking this procedure must be able to demonstrate continued competence. As the primary dangers are those of local anaesthetic toxicity, only specialist pain nursing staff trained to recognise and manage local anaesthetic toxicity will administer the infusion.

5. Equipment

5.1. Monitoring: Blood pressure, pulse, saturations, ECG and pain score every 5 minutes for the first half an hour and then subsequently at 10 minute intervals for the duration of the infusion. In addition, patients need to be assessed for symptoms of local anaesthetic toxicity such as peri-oral tingling, metallic taste, tinnitus, feeling light headed or ECG changes.

5.2. Intravenous lidocaine should be administered via a programmable syringe driver, at a fixed rate according to the protocol. It must be delivered via a sole use dedicated cannula and anti-syphon line.

5.3. A 500 ml bag of Intra-Lipid 20% must be immediately available, along with the AAGBI guideline (See Paragraph 8).

6. Procedure

6.1. Patient selection: Patients whose pain has failed to benefit from opioid and anti-neuropathic medication as assessed in outpatient clinic by Chronic Pain consultant may be considered for intravenous lidocaine infusion. Indications are limited for difficult to manage neuropathic pain such as Multiple Sclerosis pain, Trigeminal neuralgia, etc. It is not for use in generalised widespread pains such as Fibromyalgia. Caution should be applied in patients with epilepsy or cardiac disease with risk of arrhythmia and avoided in pregnant women.

6.2. Written consent is required: Lidocaine infusion for analgesia is an 'off label' use of the drug and is therefore not covered by the manufacturer's license. Doctors prescribing intravenous lidocaine for analgesia therefore assume responsibility for its use in tandem with the patient. Patients must be involved in the decision process and be made aware of the risk of local anaesthetic toxicity. These are usually in the form of feelings of light headedness, metallic taste, tinnitus and peri-oral tingling. Lidocaine has a good safety profile with slow progression of identifiable symptoms and signs prior to reaching toxic levels. Patients must be warned of the risk of toxicity which can in severe cases result in seizures, loss of consciousness and in extreme cases cardiac arrest.

6.3. Patients must avoid eating any food for 6 hours prior to infusion and drink only water up to, 2 hours prior to infusion, in case of adverse effects.

6.4. The patient must be weighed prior to writing the prescription.

6.5. The infusion must be prescribed on the inpatient prescription chart. The dose is 5 mg / Kg made up to 50 ml with 0.9% sodium chloride solution. This is then run at 25 ml per hour. If the patient develops peri-oral tingling, metal taste or feels light headed the infusion is turned down to 15 ml per hour. If symptoms persist after 15 minutes, a further reduction to 10 ml per hour may be required.

6.6. In the unlikely event of seizures, cardiac arrest or if the patient feels very unwell, the infusion must be stopped immediately. The anaesthetic consultant on call or pain consultant should be notified. Supportive care must commence immediately as detailed below, with standard life support in cardiac arrest.

6.7. Observations must be performed every 5 minutes for the first half an hour and then subsequently at 10 minute intervals for the duration of the infusion. These are to be recorded on the recovery page of the NDHT anaesthetic chart.

6.8. Patients must remain in the area for observation for 30 minutes after ceasing the infusion.

6.9. Patient must be accompanied home by a responsible adult who must remain with them for at least 12 hours. The patient may not drive for 24 hours.

7. Management of adverse effects

7.1. Metal taste / peri-oral tingling / feeling light headed

- Reduce infusion to 15mg/hr
- Check drug prescription (specific to weight) and pump infusion rate
- Contact Pain consultant
- If symptoms persist after 15 minutes reduce to 10 ml / hour
- If symptoms get worse STOP the infusion

7.2. Seizures / loss of consciousness

- Stop lidocaine infusion
- Initiate supportive care and manage the patient's airway
- If unable to contact pain consultant, fast bleep on call anaesthetic consultant

7.3. Cardiac Arrest

- Immediately initiate life support with cardio-pulmonary resuscitation (CPR)
- Stop lidocaine infusion
- Call cardiac arrest team
- Administer intra-lipid 20% 1.5 ml / kg and initiate infusion at 15 ml / kg / hour as per AAGBI guidelines on local anaesthetic toxicity
- Contact Pain consultant

8. References

- Cochrane review: Systemic administration of local anaesthetic agents to relieve neuropathic pain, 2005: http://onlinelibrary.wiley.com/doi/10.1002/14651858.CD003345.pub2/full
- Association of Anaesthetists of Great Britain and Ireland (AAGBI) guidelines on management local anaesthetic toxicity: https://www.aagbi.org/sites/default/files/la_toxicity_2010_0.pdf

9. Associated Documentation

- NDHT Anaesthetic chart
- NDHT Injectable Medicines Policy: http://www.northdevonhealth.nhs.uk/wp-content/uploads/2016/08/Injectable-Medicines-Policy-v1.4.pdf
- NDHT Resuscitation Policy: http://www.northdevonhealth.nhs.uk/wp-content/uploads/2014/02/Resuscitation-Policy-V5.0-28Oct14.pdf
- NDHT Consent Policy: http://www.northdevonhealth.nhs.uk//wp-content/uploads/2012/04/Consent-Policy-v3.3-28Aug14.pdf